WORKSHEET 4.4 : SOLVING SYSTEMS USING MATRICES

4) $x+y+z=-1$ ①
$2x+2y+3z=-7$ ②
$3x-2y+7z=-4$ ③

① × 2 $2x+2y+2z=-2$
② $2x+2y+3z=-7$
⊖ $-z=5$
$\boxed{z=-5}$ ④

① × 3 $3x+3y+3z=-3$
③ $3x-2y+7z=-4$
⊖ $5y-4z=-7$ ⑤

Sub ④ into ⑤
$5y-4(-5)=-7$
$5y+20=-7$
$5y=-27$
$\boxed{y=\frac{-27}{5}}$

sub into ①
$x+\left(\frac{-27}{5}\right)+(-5)=-1$
$x-\frac{27}{5}=4$
$5x-27=20$
$5x=47$
$\boxed{x=\frac{47}{5}}$

$\therefore$ P.O.I. $\left(\frac{47}{5}, \frac{-27}{5}, -5\right)$

5a) $\left[\begin{array}{ccc|c} 5 & -2 & 1 & 5 \\ 3 & 1 & -5 & 12 \\ 1 & -5 & 2 & -3 \end{array}\right]$ ① ② ③

b) $\left[\begin{array}{ccc|c} -2 & 1 & -3 & 0 \\ 1 & 5 & 0 & 8 \\ 0 & 3 & 2 & -6 \end{array}\right]$

c) $\left[\begin{array}{ccc|c} 0 & 4 & -3 & 12 \\ 2 & 5 & 0 & 15 \\ 4 & 0 & 6 & 10 \end{array}\right]$

6a) $x=8$
$y=-6$
$z=3$
1 solution

b) $x-6z=4$
$y+5z=-5$
$0z=0$
Infinite solution

c) $x=0$
$y=0$
$0z=1$
no solution

8a) $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ -1 & 1 & 2 & 7 \\ 2 & 1 & -1 & 4 \end{array}\right] \xrightarrow[2R_1-R_3]{R_1+R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 0 & 0 & 3 & 12 \\ 0 & -3 & 3 & 6 \end{array}\right] \xrightarrow[\frac{1}{3}R_2]{-\frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right]$

$\therefore \quad x-y+z=5 \quad ①$

$y-z=-2 \quad ②$

$\boxed{z=4} \quad ③$

sub ③ into ②

$y-(4)=-2$

$\boxed{y=2}$

sub into ①

$x-(2)+(4)=5$

$\boxed{x=3}$

∴ a unique solution at P.O.I. $(3,2,4)$

b) $\left[\begin{array}{ccc|c} 2 & 1 & -3 & 1 \\ -6 & -3 & 9 & 3 \\ 4 & 2 & -6 & 2 \end{array}\right] \xrightarrow[2R_1-R_3]{3R_1+R_2} \left[\begin{array}{ccc|c} 2 & 1 & -3 & 1 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$\therefore \quad 2x+y-3z=1$

$0y=6 \rightarrow$ not possible !!

$0z=0.$

∴ no solution.

c) $\left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ -3 & -6 & 3 & -3 \\ 2 & -4 & 2 & 4 \end{array}\right] \xrightarrow[2R_1-R_3]{3R_1+R_2} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & -12 & 6 & 4 \\ 0 & 0 & 0 & -2 \end{array}\right] \rightarrow$ not possible

∴ no solution

d) $\left[\begin{array}{ccc|c} 4 & -3 & -15 & 20 \\ 1 & 6 & 3 & 5 \\ 2 & -1 & -7 & 10 \end{array}\right] \xrightarrow[R_2]{R_1} \left[\begin{array}{ccc|c} 1 & 6 & 3 & 5 \\ 4 & -3 & -15 & 20 \\ 2 & -1 & -7 & 10 \end{array}\right] \xrightarrow[2R_1-R_3]{4R_1-R_2} \left[\begin{array}{ccc|c} 1 & 6 & 3 & 5 \\ 0 & 27 & 27 & 0 \\ 0 & 13 & 13 & 0 \end{array}\right]$

$\therefore \quad 13y+13z=0$

$z=-y, \quad y, \quad x+6y+3z=5$

$x+6y-3y=5$

$\therefore \vec{r}=(5,0,0)+t(-3,1,-1), \quad t \in \mathbb{R}$

L.O.I.

e) $3x+5y-z=1$

$-6x-10y+2z=-2$ ↙ $\times 2$ every term. ∴ same plane

$4x+2y-6z=2$ ↙ $\times 4$ every term !!

Since all these equations are "scalar multiples",

∴ they all represent the SAME equation !

f) $x-3y=1$ ①

$-2x+4y+z=1$ ②

$-2y+z=3$ ③

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 1 \\ -2 & 4 & 1 & 1 \\ 0 & -2 & 1 & 3 \end{array}\right] \rightarrow 2R_1+R_2 \left[\begin{array}{ccc|c} 1 & -3 & 0 & 1 \\ 0 & -2 & 1 & 3 \\ 0 & -2 & 1 & 3 \end{array}\right]$$

$$\rightarrow R_2 \left[\begin{array}{ccc|c} 1 & -3 & 0 & 1 \\ 0 & -2 & 1 & 3 \\ 0 & -2 & 1 & 3 \end{array}\right] \rightarrow R_2-R_3 \left[\begin{array}{ccc|c} 1 & -3 & 0 & 1 \\ 0 & -2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

① $x-3y=1$

② $-2y+z=3$

③ $0z=0 \Rightarrow$ anything !!

∴ Let $\boxed{z=t}$

② $-2y+(t)=3$

$-2y=3-t$

$\boxed{y=\frac{-3}{2}+\frac{1}{2}t}$

sub into ①

$x-3y=1$

$x-3\left(\frac{-3}{2}+\frac{1}{2}t\right)=1$

$x+\frac{9}{2}-\frac{3}{2}t=1$

$2x+9-3t=2$

$2x=-7+3t$

$\boxed{x=\frac{-7}{2}+\frac{3}{2}t}$

∴ a Line of intersection

$(x,y,z)=\left(\frac{-7}{2},\frac{-3}{2},0\right)+t\left(\frac{-3}{2},\frac{1}{2},1\right), t\in\mathbb{R}$

$$\boxed{(x,y,z)=\left(\frac{-7}{2},\frac{-3}{2},0\right)+s(3,-1,2), s\in\mathbb{R}}$$